

RUSE, Bulgaria

WMO: 155350

Lat: 43.850N

Lon: 25.950E

Elev: 148

StdP: 14.62

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.0	15.7	7.6	8.2	11.8	12.7	10.6	16.5	38.3	33.7	35.2	35.7	5.7	50	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	96.0	72.3	92.5	71.5	89.6	70.4	75.0	90.1	73.4	88.2	71.9	86.2	7.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.1	111.5	83.5	68.4	105.0	80.8	67.0	100.0	78.8	38.5	90.1	37.0	88.0	35.7	86.4	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
32.2	27.3	22.2	DB	5.9	102.2	6.8	4.0	1.0	105.1	-3.0	107.5	-6.9	109.7	-11.8	112.7	(4)
			WB	5.5	78.8	6.8	2.7	0.6	80.8	-3.4	82.4	-7.2	83.9	-12.2	85.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.5	31.9	36.8	46.3	56.2	66.1	72.8	76.7	76.5	67.1	55.9	44.7	34.3
	DBStd	17.57	9.32	10.16	8.76	6.90	6.39	5.91	5.27	5.48	6.57	8.10	8.16	9.04
	HDD50	1871	562	378	178	25	0	0	0	0	0	36	200	492
	HDD65	4658	1025	791	582	271	65	9	1	1	53	298	609	953
	CDD50	3895	2	8	62	212	500	684	827	822	513	219	42	4
	CDD65	1204	0	0	1	8	99	242	363	358	116	16	1	0
	CDH74	11758	0	0	7	92	895	2318	3760	3632	925	127	2	0
	CDH80	4974	0	0	0	21	258	937	1753	1688	291	26	0	0
Wind	WSAvg	7.6	7.3	8.4	9.4	9.0	8.0	7.6	7.3	6.7	7.1	6.5	7.0	7.3
Precipitation	PrecAvg	24.2	1.7	1.3	1.7	1.8	2.6	2.6	2.4	1.8	2.4	2.0	1.8	2.0
	PrecMax	39.1	4.5	4.1	4.1	4.8	6.9	6.6	7.9	5.3	7.8	4.9	4.5	6.5
	PrecMin	13.1	0.1	0.0	0.1	0.1	0.6	0.5	0.1	0.1	0.0	0.0	0.0	0.1
	PrecStd	6.1	1.0	0.8	0.9	0.9	1.6	1.4	2.0	1.5	2.0	1.3	1.1	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.0	65.4	73.7	82.8	90.8	96.8	101.2	100.8	92.0	83.6	70.8	61.8
		MCWB	49.1	52.6	55.8	64.1	69.1	73.3	73.1	72.9	68.9	66.1	58.8	52.9
	2%	DB	52.9	59.7	69.2	75.7	86.1	92.2	96.4	95.7	87.0	77.3	66.1	54.6
		MCWB	46.4	49.9	54.0	60.0	67.3	72.1	72.0	72.7	67.4	64.1	57.2	48.9
	5%	DB	48.1	55.0	64.8	71.8	82.6	88.8	92.5	92.2	82.9	72.1	60.6	49.4
		MCWB	43.8	47.6	52.5	57.9	65.8	71.0	71.5	71.5	66.4	61.4	54.4	45.4
	10%	DB	43.8	50.0	60.4	68.3	79.1	85.7	89.1	88.9	79.4	68.0	56.4	45.5
		MCWB	40.7	44.6	50.0	55.8	64.4	69.4	70.4	70.1	64.9	59.0	51.7	42.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.1	53.6	58.4	65.4	72.5	75.6	76.4	77.6	71.7	68.0	60.1	53.1
		MCDB	56.2	62.8	69.3	80.3	85.5	90.5	94.0	93.1	84.4	80.7	69.1	60.5
	2%	WB	47.0	50.5	54.9	61.5	69.3	73.8	74.6	75.3	69.3	64.7	57.5	49.2
		MCDB	51.8	58.7	66.9	73.7	82.6	89.0	90.7	89.6	83.6	75.4	64.7	54.2
	5%	WB	43.8	47.8	52.9	59.2	67.1	72.0	73.0	73.2	67.5	62.3	55.1	45.9
		MCDB	47.8	54.3	63.8	69.4	79.4	86.5	88.2	87.4	80.3	70.7	60.2	48.8
	10%	WB	40.9	44.9	50.5	57.1	65.2	70.2	71.7	71.4	65.7	59.9	52.2	43.0
		MCDB	43.7	49.8	59.9	66.1	76.8	83.5	85.8	85.8	76.9	67.1	55.9	45.3
Mean Daily Temperature Range	5% DB	MDBR	9.7	13.3	16.8	18.4	20.2	20.5	20.8	20.8	18.7	16.5	12.5	9.9
		MCDBR	16.2	21.7	23.8	24.2	25.7	25.4	26.4	25.9	24.8	23.0	19.8	16.6
		MCWBR	11.2	13.7	13.2	12.2	11.9	10.5	9.9	9.7	10.4	12.2	12.2	11.6
	5% WB	MCDBR	15.4	20.6	22.6	21.5	23.1	23.7	23.2	22.9	22.1	21.5	18.3	15.6
		MCWBR	10.9	13.4	13.1	11.6	12.2	10.8	9.8	10.3	10.8	11.9	11.5	11.4
Clear-Sky Solar Irradiance	taub	0.351	0.384	0.421	0.461	0.435	0.444	0.453	0.467	0.433	0.406	0.377	0.355	
	taud	2.263	2.189	2.139	2.051	2.201	2.211	2.192	2.151	2.211	2.261	2.292	2.277	
	Ebn at Noon	237	249	256	255	266	264	259	251	249	239	226	223	
	Edh at Noon	28	36	43	50	45	44	45	45	39	33	27	25	
All-Sky Solar Radiation	RadAvg	462	714	1081	1493	1855	2025	2056	1840	1337	867	495	379	
	RadStd	70	99	106	158	145	144	140	104	104	96	67	52	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (52 neighbors)		+0.81	N/A	N/A	N/A	N/A	N/A	N/A	-214	+113	N/A

Nomenclature: See separate page